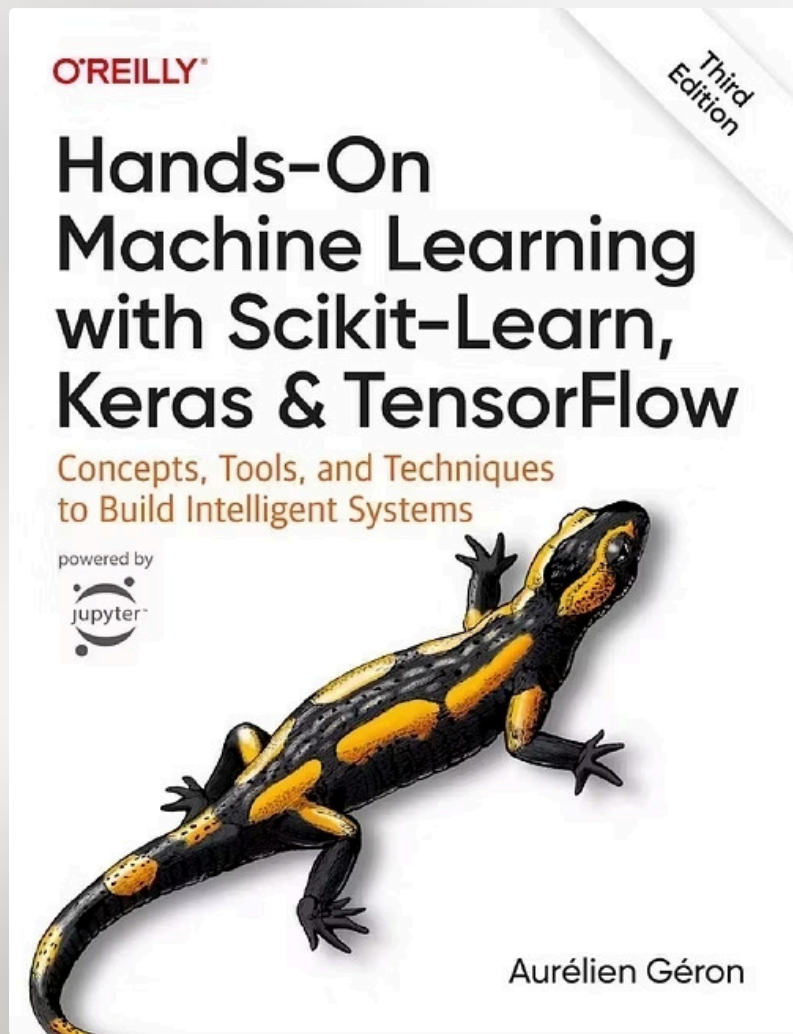




Machine Learning

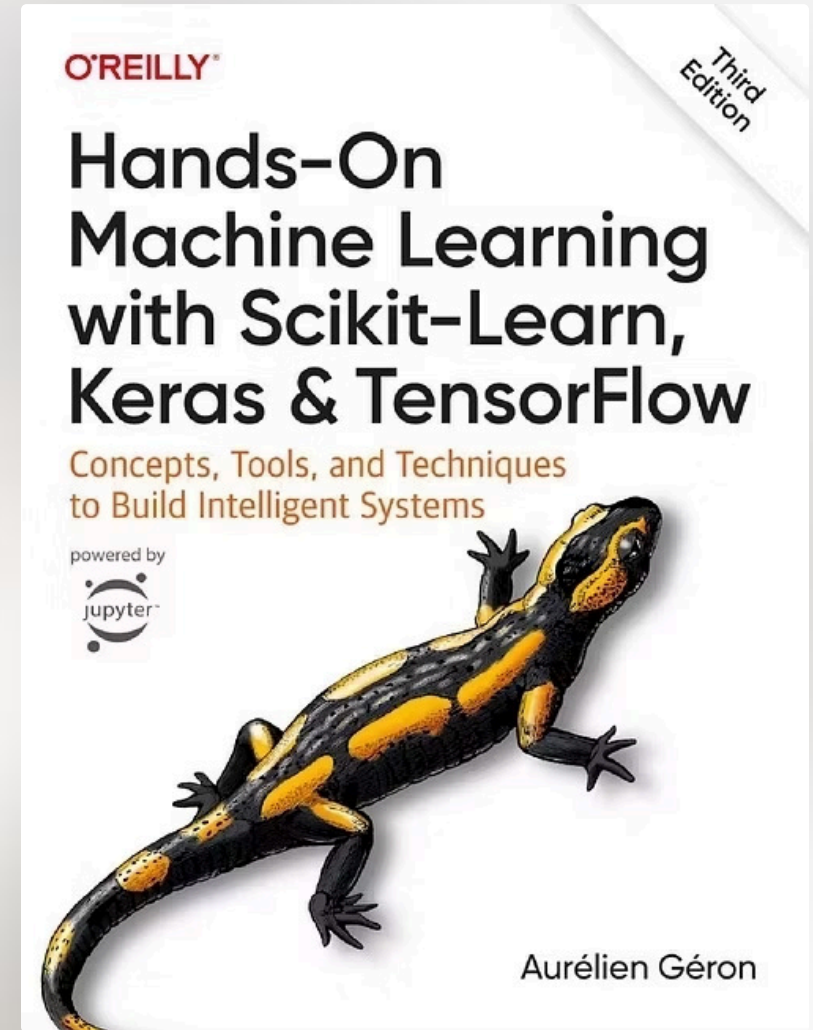
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/04/2025

Machine Learning

Chapter 1 – The Machine Learning Landscape





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1: Supervision during training**
 - **Supervised**, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement **Learning**
 - **Classification criteria #2: Approach to Learning Method**
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3: Learning Method**
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Classification criteria #1: Supervision during training

Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning

Supervised Learning: Classification Task

In supervised learning, the training set you feed to the algorithm includes the desired solutions, called labels.

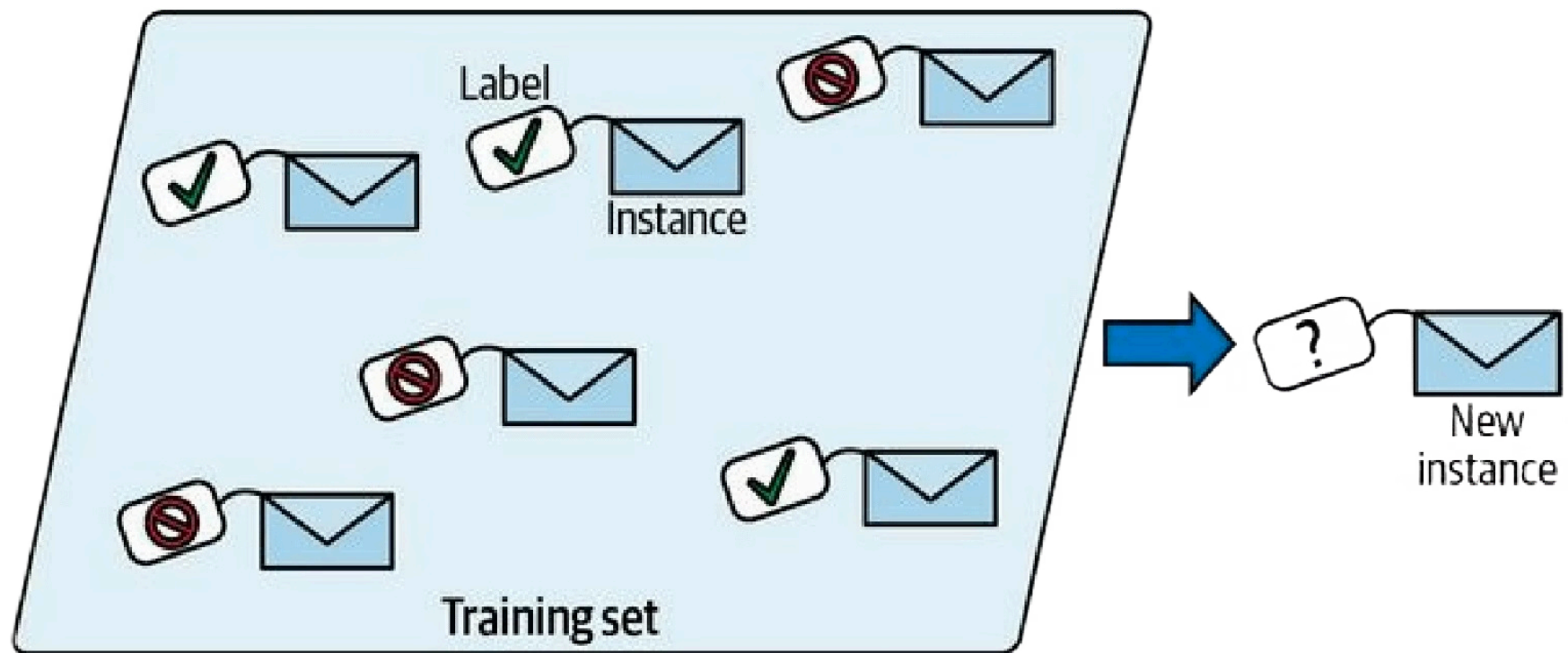


Figure 1-5. A labeled training set for spam classification (an example of supervised learning)

Supervised Learning: Regression Task

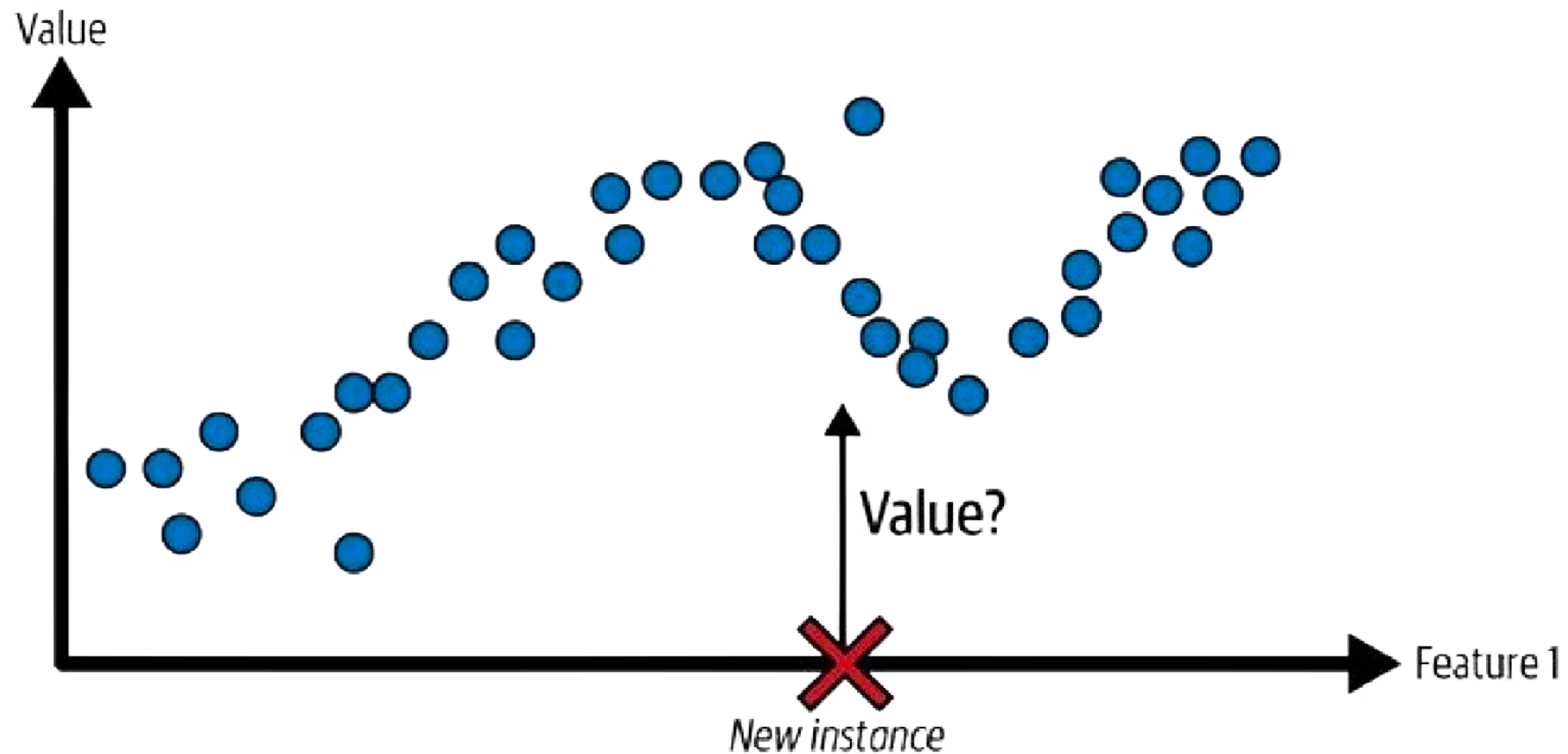


Figure 1-6. A regression problem: **predict a value, given an input feature** (there are usually multiple input features, and sometimes multiple output values)



Note that some regression models can be used for classification as well, and vice versa.

For example, logistic regression is commonly used for classification, as it can output a value that corresponds to the probability of belonging to a given class (e.g., 20% chance of being spam).



Supervised Learning Terminology

Target vs. Label

Features vs. Predictors vs. Attributes

Target vs. Label in ML

- **Target (Regression):**
 - More commonly used in **regression tasks**
 - Represents the **continuous value** the **model predicts**
 - **Example:** Predicting house prices based on various features
- **Label (Classification):**
 - More commonly used in **classification tasks**
 - Represents the **class** or **category** the **model assigns**
 - **Example:** Classifying emails as "**spam**" or "**ham**"



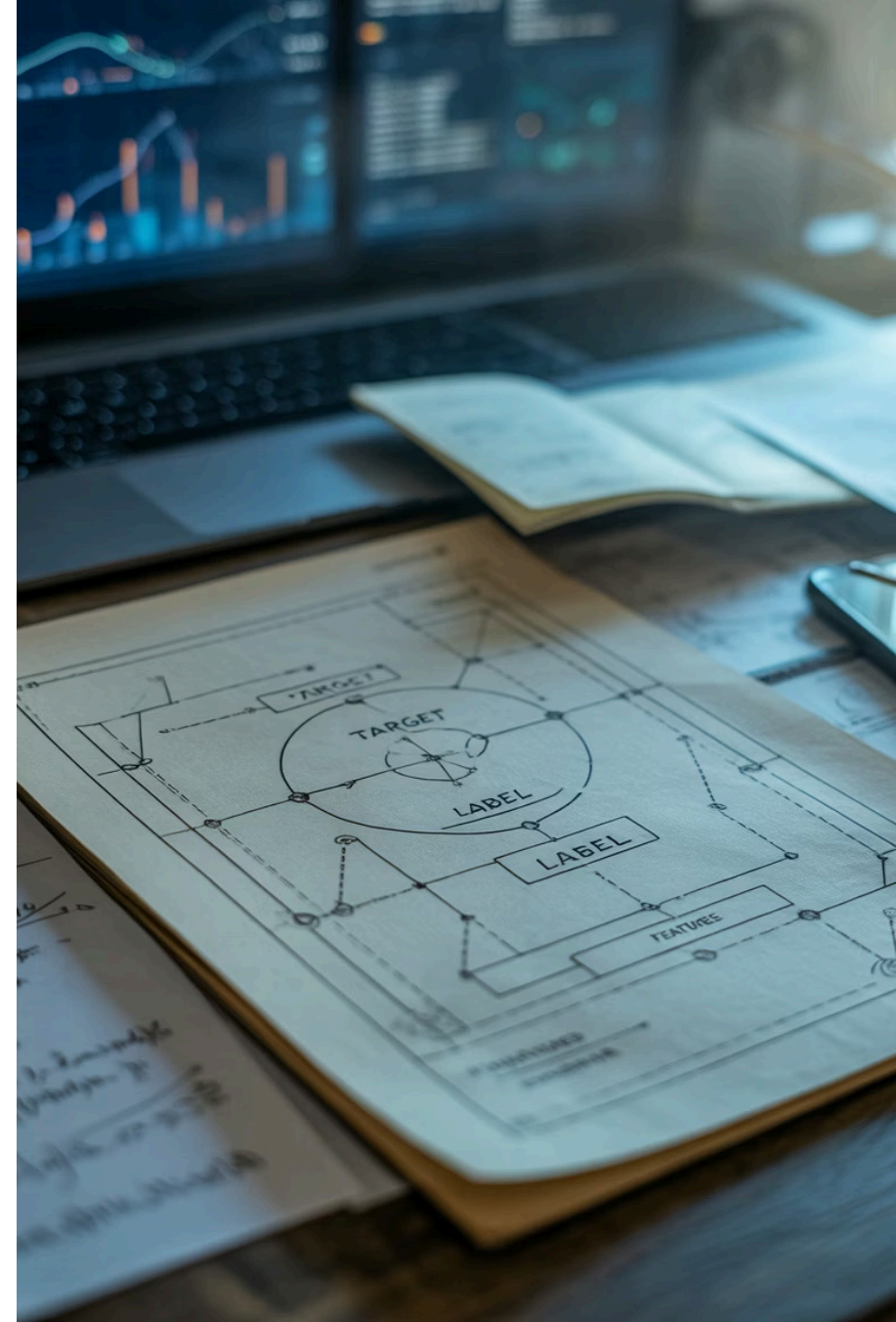


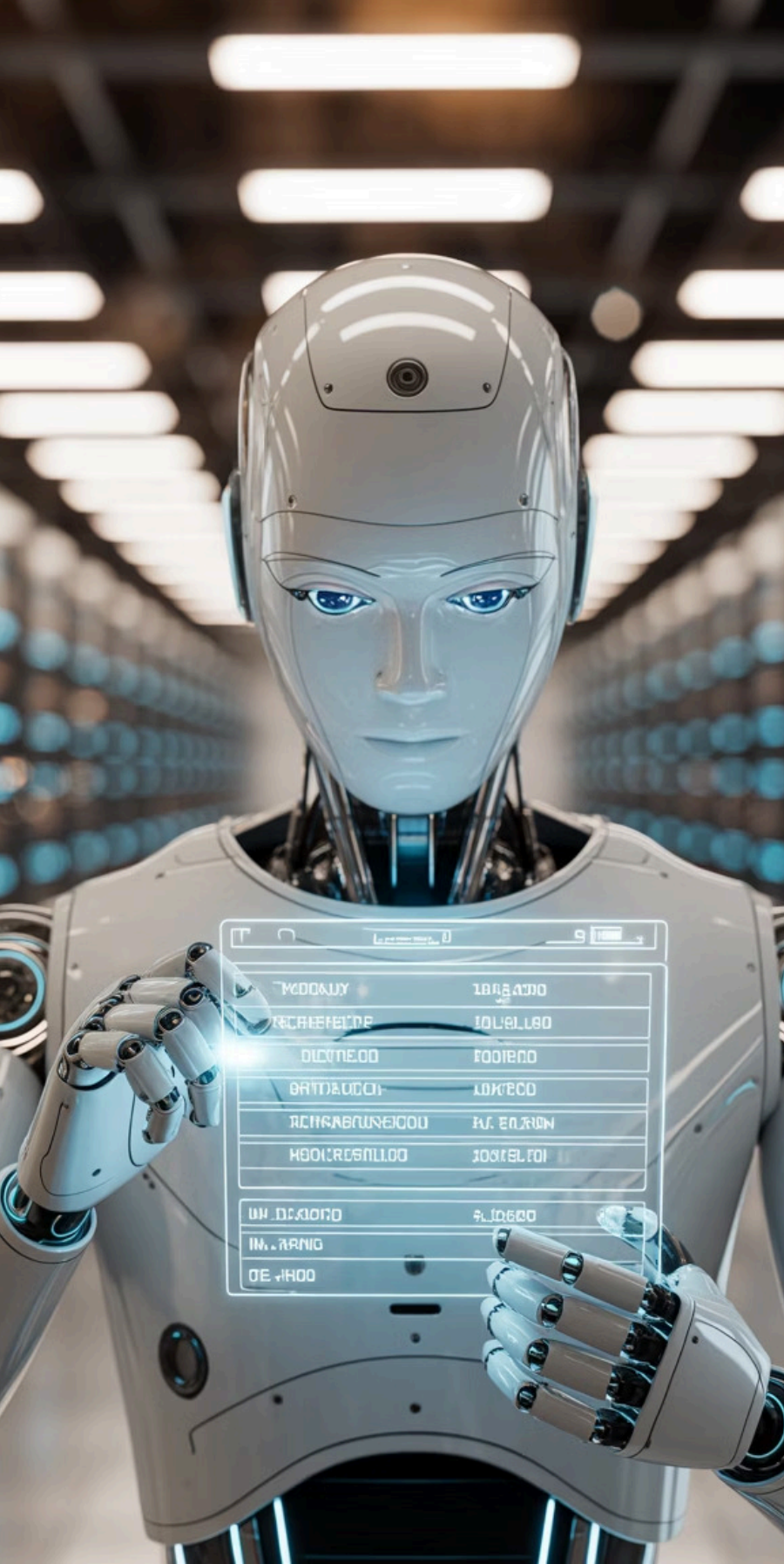
Features vs. Predictors vs. Attributes in ML

- **Features:**
 - Sometimes called **predictors** or **attributes**
 - Represent the **input variables** used by the **model** to make **predictions**
 - **Example:** Age, income, and education level may be features in a model predicting loan approval

Summary of Terms

- **Target:**
 - Used mainly in **regression tasks**, refers to the **predicted continuous value**
- **Label:**
 - Used mainly in **classification tasks**, refers to the **predicted class or category**
- **Features/predictors/attributes:**
 - **Input variables** used to train the model





Supervised Learning Summary



Labeled data

Models are trained on labeled data (inputs with corresponding outputs)



Typical Supervised Learning tasks

Classification, Regression



Classification: Predicting discrete labels

For example, spam vs. ham



Regression: Predicting continuous values

For example, house price based on features like area, location

Next: Unsupervised Learning



Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts